

# Presentation

Data Structures

2025-2026



Universidad Pública de Navarra  
Nafarroako Unibertsitate Publikoa

- **Ugaitz Amozarrain**
- Department of Statistics, Computer Science and Mathematics
- *Las Encinas* building, basement, office 9002
- Email: [ugaitz.amozarrain@unavarra.es](mailto:ugaitz.amozarrain@unavarra.es)

# Office Hours

- Monday: 16:00–18:00
- Wednesday: 11:00–13:00

## Notes

Office hours are held in my office unless stated otherwise. Please contact me in advance to agree on a time slot and avoid overcrowding.

# Course Schedule

## Theory and Lab Lectures

Monday 12:00–14:00

Theory — Classroom A-020

Thursday 12:00–14:00

Lab — Classroom A-331

# Different problems need different ways of organizing data

- Searching millions of elements → Trees
- Undo/redo in an editor → Stacks
- Task scheduling → Queues
- File compression → Huffman Coding
- Social networks → Graphs

# Semester Timeline (Overview)

- **Weeks 1–3:** Python review, Sparse matrices, OOP, file handling
- **Weeks 4–6:** Lists, stacks, queues, dequeues
- **Weeks 7–9:** Linked lists
- **Weeks 10–11:** Recursion and complexity, binary trees
- **Weeks 12–15:** Trees, graphs, Huffman coding

# Evaluation Criteria

| <b>Evaluation Method</b>      | <b>Weight</b> | <b>Retake</b> |
|-------------------------------|---------------|---------------|
| Theoretical / Practical Exams | 50%           | Yes           |
| Deliverables                  | 50%           | Yes           |

# Exam Details

- Exam value: 5 points
- Practical exam performed on computer
  - You will be able to test your code
  - No access to the internet
- Mandatory to pass the exam to pass the course
- Retake exam counts for 100% of the final grade



# Deliverables

- Deliverables value: 5 points
- Weekly lab assignments
  - All submissions are checked for correctness
- 2 bigger projects that will contain a practical part and a report

| <b>Lab Project</b>        | <b>Delivery/Correction</b> | <b>Max points</b> |
|---------------------------|----------------------------|-------------------|
| Linked Lists              | Correction                 | 2                 |
| Huffman Coding            | Correction                 | 2                 |
| Sparse Matrices           | Delivery                   |                   |
| Working with Files        | Delivery                   |                   |
| Postfix Calculator        | Delivery                   |                   |
| Palindromes               | Delivery                   |                   |
| Balanced Parentheses      | Delivery                   |                   |
| Conway's game of Life     | Delivery                   |                   |
| Recursion Exercises       | Delivery                   |                   |
| Constructing Binary Trees | Delivery                   |                   |

# Exams

- 25/05/2026 Final exam (16:00)  
Covers all course contents
- 17/06/2026 Retake exam (16:00)  
Covers all course contents

# Use of Generative AI

## Careful with AI tools

The intention of this course is for you to learn how to create and use data structures. Do not let Generative AI do the work for you.